Tribhuvan University

Faculty of Humanities and Social Science

A PROJECT REPORT FOR

"Bus Ticketing System using Closest Pickup and Content Based Algorithms"

Submitted to

Department of Computer Application

Advanced College of Engineering and Management

Kalanki, Kathmandu

In partial fulfillment of the requirements for the bachelor's in computer application

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Tribhuvan University

Faculty of Humanities and Social Science

Advance College of Engineering and Management

Kalanki, Kathmandu

Bachelor in Computer Application (BCA)

SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project prepared under my supervision by **Rubek Maharjan and Stephen G. Shrestha** entitled **"Bus Ticketing System using Closest Pickup and Content Based Algorithms"** in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

……………………………………

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SUPERVISOR

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ADVANCED COLLEGE OF ENGINEERING AND MANAGEMENT

LETTER OF ACCEPTANCE

This is to certify that this project prepared by **"Rubek Maharjan and Stephen G. Shrestha"** entitled **"MahaBus"** in partial fulfillment of the requirements for the 6th semester of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the defense project.

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Er. Sakriya Maharjan

Project Supervisor

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ABSTRACT

The Bus Ticketing System is a comprehensive platform designed to streamline the booking and management of bus tickets. It allows users to search for bus routes, book tickets, and access travel schedules, ensuring a convenient and efficient booking experience. The system features a dual-panel architecture, with distinct capabilities for users and administrators. Users can easily view available routes, select preferred seats, and make payments, while administrators can manage schedules, allocate buses, and monitor ticket sales and occupancy rates. This organized approach enhances operational efficiency and minimizes manual errors, providing a seamless experience for all stakeholders. The system's architecture supports extensive data collection and processing, ensuring efficient management of both user interactions and administrative tasks. The software aims to improve the ticketing process by offering an intuitive interface and integrated communication features. Future developments will focus on advanced data analytics and machine learning techniques to optimize route planning, enhance user experience, and provide detailed insights into travel patterns and preferences. This innovative technology offers a forward-thinking solution for improving the organization and execution of bus ticketing in an increasingly digital world.

Keywords: HTML, CSS, JavaScript, PHP, Content-Based Filtering, SQL, Map Integration, UI/UX

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Yours Sincerely,

Rubek Maharjan

Stephen G. Shrestha

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LIST OF ABBREVIATIONS

CASE	Computer Aided Software Engineering
CSS	Cascading Style Sheet
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
HTML	Hypertext Markup Language
JS	JavaScript
MySQL	Microsoft Server Structured Query Language
PHP	Hypertext Preprocessor
SQL	Structured Query Language
UI	User Interface
RDMS	Relational Database Management System

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CHAPTER 1

INTRODUCTION

1.1 Introduction

In the current era of digital transformation, public transportation systems are undergoing significant changes to enhance service efficiency, accessibility, and passenger satisfaction. One critical aspect of these advancements is the modernization of bus ticketing systems. Traditional ticketing methods, often characterized by long queues, paper tickets, and manual processing, are increasingly becoming obsolete. [1]

The proposed Bus Ticketing System aims to address the challenges faced by both passengers and transportation providers. For passengers, the system will offer an intuitive platform to easily book, manage, and validate bus tickets through website. The system will contribute to sustainable mobility by reducing the reliance on paper tickets and supporting the development of smart cities.

1.2 Problem Statement:

Booking bus tickets in Nepal is often complicated and time-consuming, causing frustration for both passengers and bus operators. Passengers face challenges such as unclear booking processes and limited access to information, while bus companies struggle with managing bookings efficiently.

This project aims to develop a user-friendly Bus Ticket Booking System specifically for Nepal. The system will simplify ticket reservations for passengers, ensuring a smoother booking experience. For bus operators, it will provide tools to manage schedules effectively and improve customer service, ultimately enhancing the overall efficiency of the bus transport system in Nepal.

1.3 Objectives:

- To create simple platform to book bus tickets efficiently.
- To develop tools for bus operators to manage schedules, seats, and routes.

1.4 Scope & Limitation

1.4.1 Scope

The objective of this project is to design, develop, and deploy a comprehensive online platform for booking bus tickets. The system will serve as a centralized hub where users can search for bus routes, compare schedules, select seats, and complete their ticket purchases. The platform will provide detailed bus information, real-time seat availability, and payment options, delivering a seamless experience for both passengers and bus operators.

The project will involve the following key components:

User Registration and Authentication: Users will be able to create accounts, log in, and manage their profiles. They can also save routes, track bookings, and receive notifications about upcoming trips, special offers, and updates.

Bus Schedule and Route Listings: The system will provide a comprehensive database of bus routes and schedules. Users will be able to search for buses by departure and destination locations, dates, times, and bus type (luxury or regular). Detailed information on each route, including departure times, travel durations, stops, and end destination.

Ticket Booking and Payment: Users will be able to select their desired bus, route, time, and seats and proceed to book their tickets. The platform will support multiple payment options, including credit/debit cards, digital wallets, and online banking. Users will have the option to modify or cancel their bookings within the system's cancellation policies.

1.4.2 Limitation

Dependence on Internet Connectivity: The system relies heavily on internet access for users to book tickets, view schedules, and check real-time seat availability. Limited or no internet access could prevent passengers from using the platform, especially in rural areas or during travel.

Real-Time Data Sync Issues: The system's accuracy depends on real-time updates from bus operators. Delays or failures in syncing data (seat availability, cancellations) could lead to inaccurate information being displayed, resulting in booking errors or double bookings.

Scalability Challenges: As the number of users, routes, and transactions grows, the system may face scalability challenges. If not built with scalability in mind, it could experience slowdowns or crashes during periods of high demand, such as holidays or peak travel times.

Refund and Cancellation Delays: While the system automates ticket cancellations and refunds, delays in processing refunds due to banking systems or payment gateway issues could lead to dissatisfaction among passengers, especially in urgent situations.

1.5 Report Organization

Chapter 1 introduces the system, we outline its objectives, limitations, and the compelling reasons that drove its development. By providing a concise overview, we aim to offer a clear understanding of the system's purpose and the boundaries it operates within.

Chapter 2 summarizes system management, various applications have been developed with intriguing features.

Chapter 3 encompass various aspects, including functional and non-functional requirements, feasibility analysis, ER diagram, DFD, system design, system architecture, database schema, and interface design.

Chapter 4 talks about various tools that were used in the system development process to facilitate implementation. Detailed implementation procedures were followed, including thorough testing to assess the system's performance and functionality.

Chapter 5 highlights a brief summary of lessons learnt, outcome and conclusion of the whole project and explains what has been done and what further improvements could be done.

CHAPTER 2

BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

In many countries, the bus transportation system operates informally, with no centralized or standardized platform for ticketing, scheduling, or managing bus routes. Often, passengers have to rely on physical ticket counters, bus terminals, or local agents to book tickets, leading to inconsistent pricing, lack of transparency, and frequent overbooking. Furthermore, pricing is often non-standardized, with rates varying based on demand or the operator's discretion.

Due to these challenges, several bus ticketing websites and mobile applications have emerged in the market. Some platforms favor larger bus operators, leaving smaller regional services underrepresented, while others may not provide users with the flexibility to modify or cancel bookings easily. [2]

In response to this situation, we are developing a comprehensive bus ticketing website that addresses the needs of both passengers and bus operators. The goal is to provide a user-friendly platform that offers a transparent, real-time view of bus schedules, seat availability, and pricing. The system will also allow bus operators to list and manage their routes, prices, and services in a competitive environment. By balancing the needs of passengers, operators, and local businesses, this platform aims to standardize the bus ticketing process, making it accessible, efficient, and equitable for all users involved.

2.2 Literature Review

A study by Sharma explores how user experience impacts the adoption and satisfaction of online bus ticket booking systems. Findings highlight that interface design, ease of use, and customer support significantly influence user behavior and system effectiveness, crucial for enhancing service quality and user satisfaction. [3]

According to a report by World Bank, technological innovations such as real-time data integration, mobile applications, and AI-driven analytics are transforming transportation management systems. These innovations improve operational efficiency, scheduling accuracy, and customer service within bus ticket booking platforms, fostering a more responsive and reliable travel experience. [4]

Research by J.D. Power examines customer satisfaction and loyalty in online travel booking, including bus ticket reservations. The study identifies key drivers such as service quality, price transparency, booking convenience, and post-purchase support, influencing customer retention and brand loyalty in competitive travel markets. [5]

MarketsandMarkets' report explores the integration of Internet of Things (IoT) devices, such as smart ticketing systems and vehicle trackers, in public transportation. IoT platforms and edge computing enable real-time data processing, enhancing operational efficiency and passenger experience. [6]

McKinsey & Company's analysis explores the application of AI and machine learning algorithms in personalizing travel recommendations and optimizing pricing strategies. AI-driven chatbots, predictive analytics, and dynamic pricing models enhance customer engagement and revenue generation in bus ticketing platforms. [7]

Studies by UC Berkeley assess the impact of ride-sharing services (e.g., Uber, Lyft) on bus travel demand and service utilization. Integration with multi-modal transportation apps and shared mobility solutions offer commuters flexible travel options and influence modal shift behaviors in urban areas. [8]

Security measures in bus ticketing systems include HTTPS protocol for secure data transmission, SSL/TLS encryption for data protection, and OAuth/JWT for secure authentication. Regular security audits and vulnerability assessments are conducted to detect and mitigate threats, complemented by firewall configurations and intrusion detection systems (IDS) to ensure compliance with regulatory standards like PCI-DSS, maintaining user trust and system integrity. [9]

According to Mbarika manual systems were prone to errors, with issues like overbooking or inconsistent pricing being common. Moreover, passengers lacked access to real-time information, such as bus schedules, delays, or cancellations. [10]

Grob and Bernon discuss how the introduction of digital ticketing revolutionized the transportation sector, as users could book tickets online or through mobile devices, reducing the reliance on physical ticket counters and improving operational efficiency. [11]

Shiroor analyze the development of online ticketing platforms, noting that the integration of web-based and mobile applications has become a game-changer for public transport. These platforms provide users with the ability to search for available buses, view schedules, select seats, and pay for tickets online. [12]

Kaur and Sandhu, online systems offer bus operators the ability to optimize route planning and dynamically adjust prices based on demand. [13]

According to Kumar, allowed users to book tickets on the go, view seat layouts, and receive real-time alerts. The convenience provided by these apps enhanced the user experience and widened the customer base, as passengers no longer needed access to desktop computers to make bookings. [14]

Sasidharan highlight the issue of digital divide, where passengers in rural areas or regions with poor internet connectivity may struggle to use online systems. In these cases, the reliance on manual or agent-based ticketing still persists, which can limit the reach of digital platforms. [15]

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

3.1.1 Requirement Analysis

The requirement analysis for a Bus Ticket Booking System in Nepal involves identifying and documenting the functional and non-functional requirements necessary to develop a comprehensive and efficient system. This analysis aims to ensure that the system meets the needs of users, bus operators, and other stakeholders.

i. Functional Requirement

There are various basic as well as secondary functional requirements for a bus ticketing website. Some of them are user registration and authentication, manages buses, view buses, book seats, view ticket etc. User authentication should ensure that only authorized users can access certain features and information. Similarly, users should be able to search and view for buses based on various criteria, such as location and specific features.

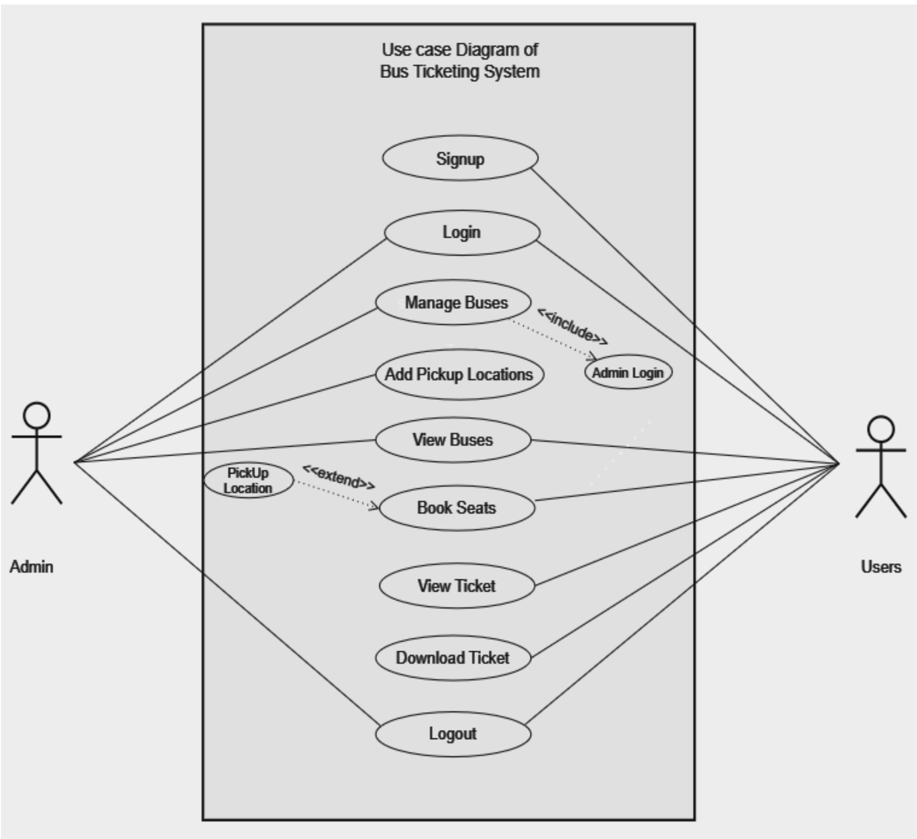

Figure 3. 1 Use Case Diagram of Bus Ticketing System

In the above use case diagram, there are two actors (admin and users). The actors can login and logout. The admin initiates the manage buses, add pickup location and view buses. User's initiates view buses, book seats, view ticket and download ticket.

ii. Non-Functional Requirement

Performance, usability, user experience and security are important non-functional requirements for a website. The website should have fast loading times to ensure a smooth user experience. Content and features should adapt to smaller screens without loss of functionality or readability. The website should employ robust security measures to protect user data and prevent unauthorized access. User login and authentication processes should be secure and follow best practices.

3.1.2 Feasibility Analysis

Feasibility study is a process of analyzing technical, operational and economic viability of a project. It ensures that the proposed system aligns with the team's goals, resources, and capabilities.

i. Technical

The technology needed to build the Bus Ticket Booking System is readily available. We can use existing platforms and tools to develop both web and mobile applications. Reliable internet connectivity and smartphone usage are growing in Nepal, making it feasible to implement and use this system.

HTML, CSS and JavaScript will be the frontend technologies for this website. Similarly, its backend will be written on PHP and MySQL will be used as database. Additionally, the website will require regular maintenance and updates to ensure it remains up-to-date with the latest technologies and security standards.

ii. Operational

The system will simplify the bus ticket booking process for users and bus operators. Users will find it easier to search for and book tickets online, reducing the need for physical ticket counters. Bus operators will benefit from automated booking management, reducing manual errors and improving efficiency. The system will include customer support and notifications, enhancing user satisfaction.

iii. Economic

Developing the system requires an initial investment in software development, server hosting, and marketing. However, the costs can be offset by the revenue generated from booking fees and partnerships with bus operators. The convenience and improved service quality will attract more users, leading to increased bookings and profitability over time. Overall, the long-term financial benefits outweigh the initial costs.

3.1.3 Data Modeling (ER Diagram)

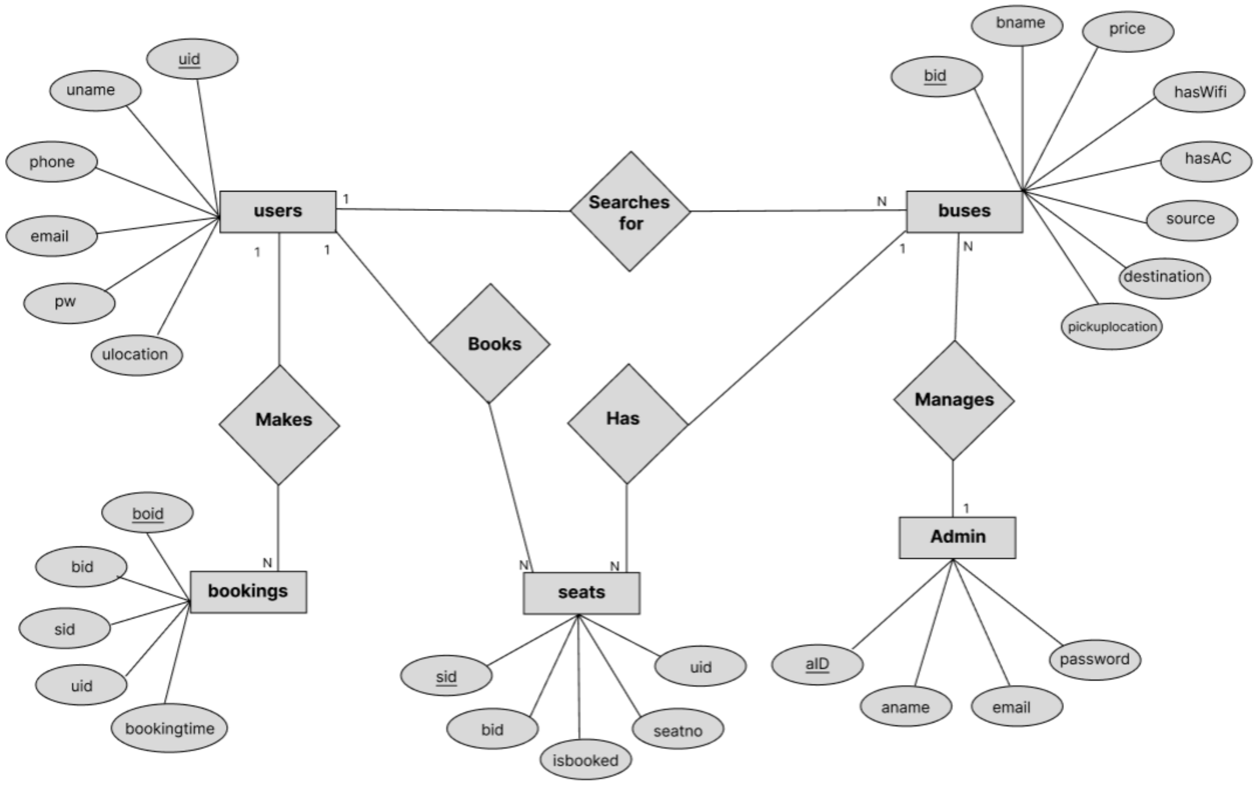

Figure 3. 2 ER Diagram of Bus Ticketing System

In above ER Diagram, users, buses, bookings, admin and seats are the entities which are represent by the rectangle. ulocation, phone, email, pw, uname, source, destination etc are the attributes of the entities. Eclipse is used to represent an attribute. Books, Searches for, manages, makes and has are the relationship which is used to describe the relation between entities. Diamond is used to represent the relationship.

3.1.4 Process Modeling (DFD)

3.1.4.1 Context Diagram

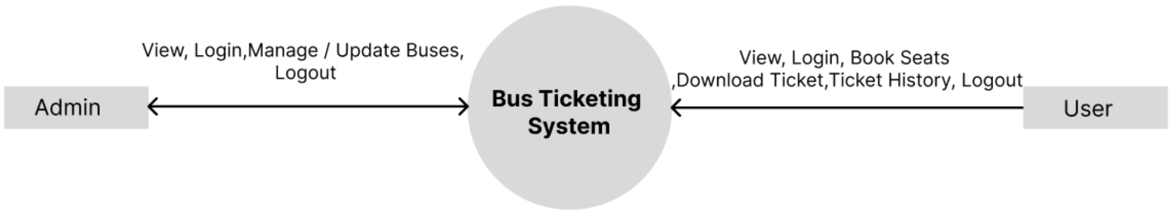

Figure 3. 3 Context Modeling of Bus Ticketing System

Level 0 Data Flow Diagram (DFD) provides a high-level overview of the project functionality, illustrating the interactions between external entities and the system itself.

In the diagram above, the external entity are admin and user. The entities can view system as well as login and logout. The entity (admin) can manage & update the buses whereas entity (user) can book seats, download ticket and view the past history.

3.1.4.2 Level 1 DFD

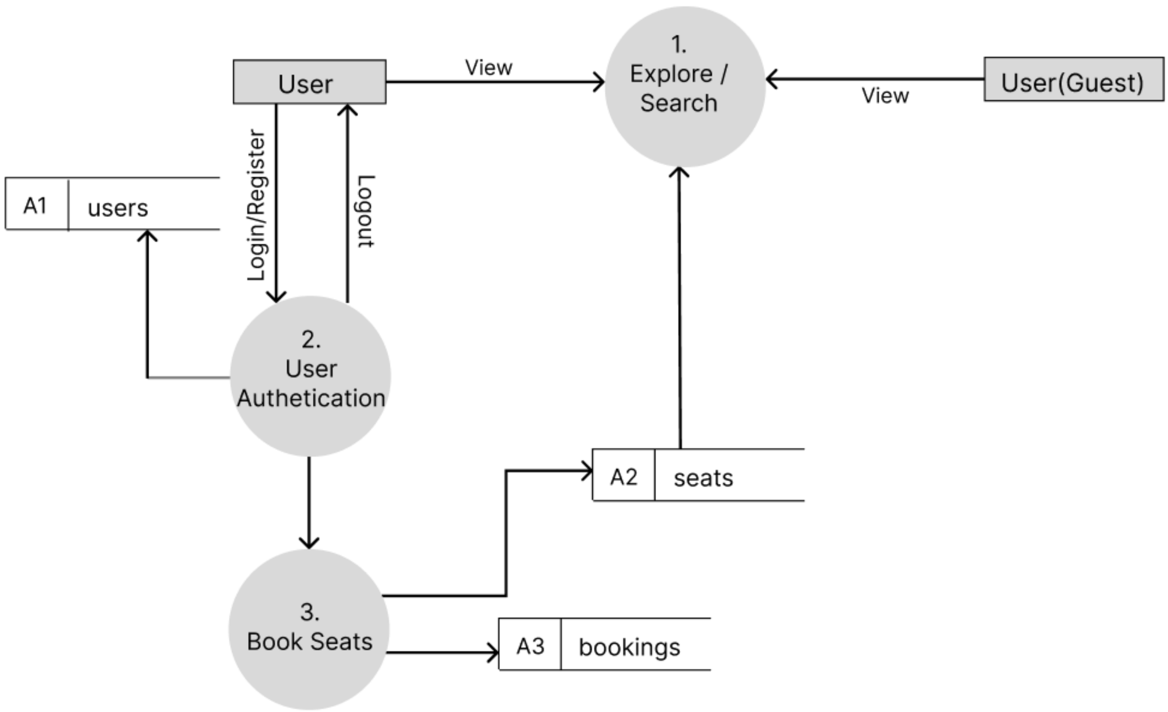

Figure 3. 4 Level 1 DFD of Bus Ticketing System

The above diagram illustrates level 1 DFD diagram where the user firstly opens the website. Then the user can register their account to use all the facilities that are provided by the web application. After login, user can view buses and book the seats on this website. The data entered by the user are stored in different databases. The user(guest) can use the explore option based on their needs.

3.1.4.3 Level 2 DFD

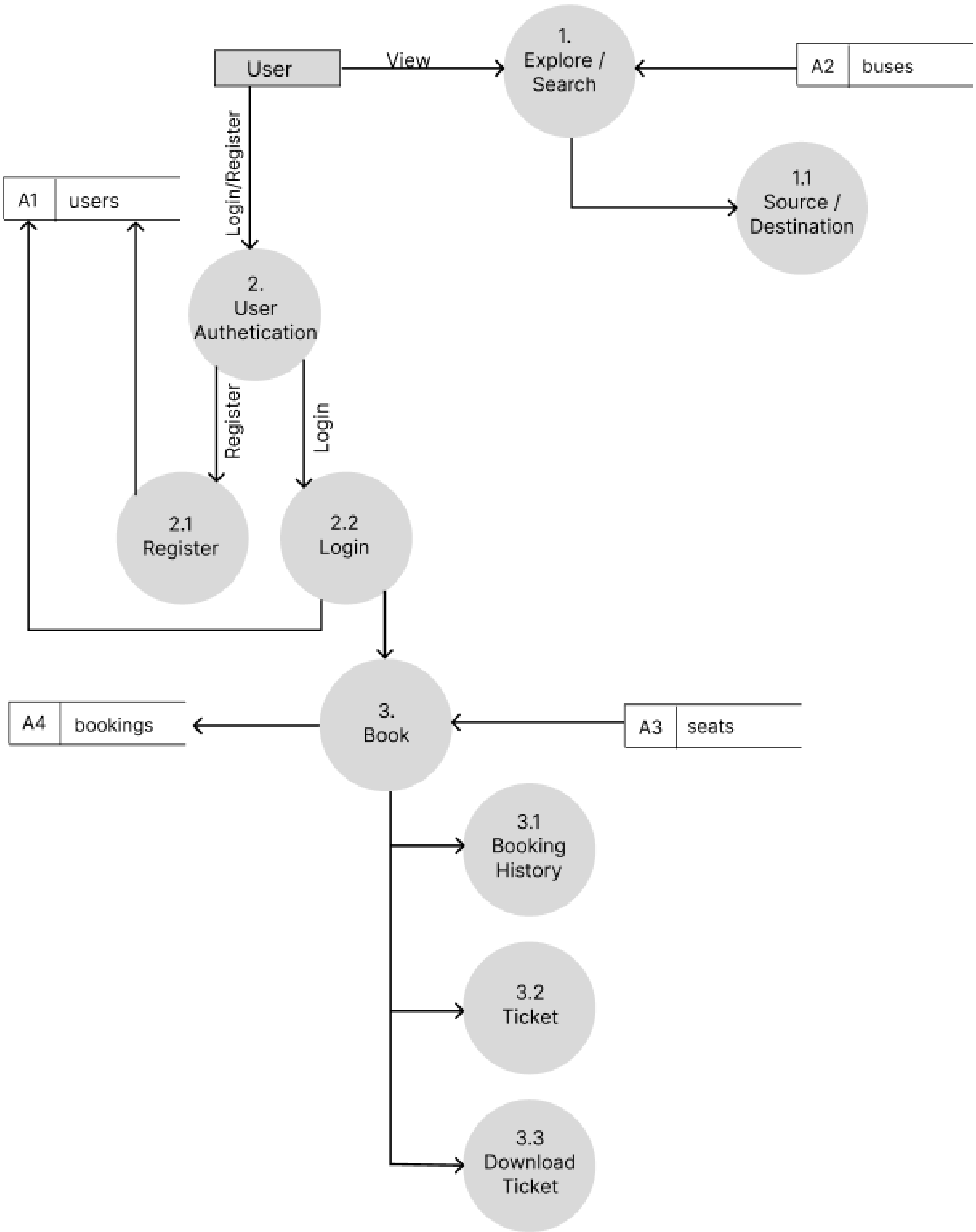

Figure 3. 5 Level 2 DFD of Bus Ticketing System

The above diagram illustrates level 1 DFD diagram where the user firstly opens the website. If the user is using the system then the system asks for user authentication. If the user has their account then they can use all the facilities that are provided by the web application. Each user's details are stored in database. After login, user can view buses, book the seats and watch the past booking history on this website. The user can also download the ticket easily on their devices. The data enter by the users are stored in different database

3.1.4.4 Physical DFD

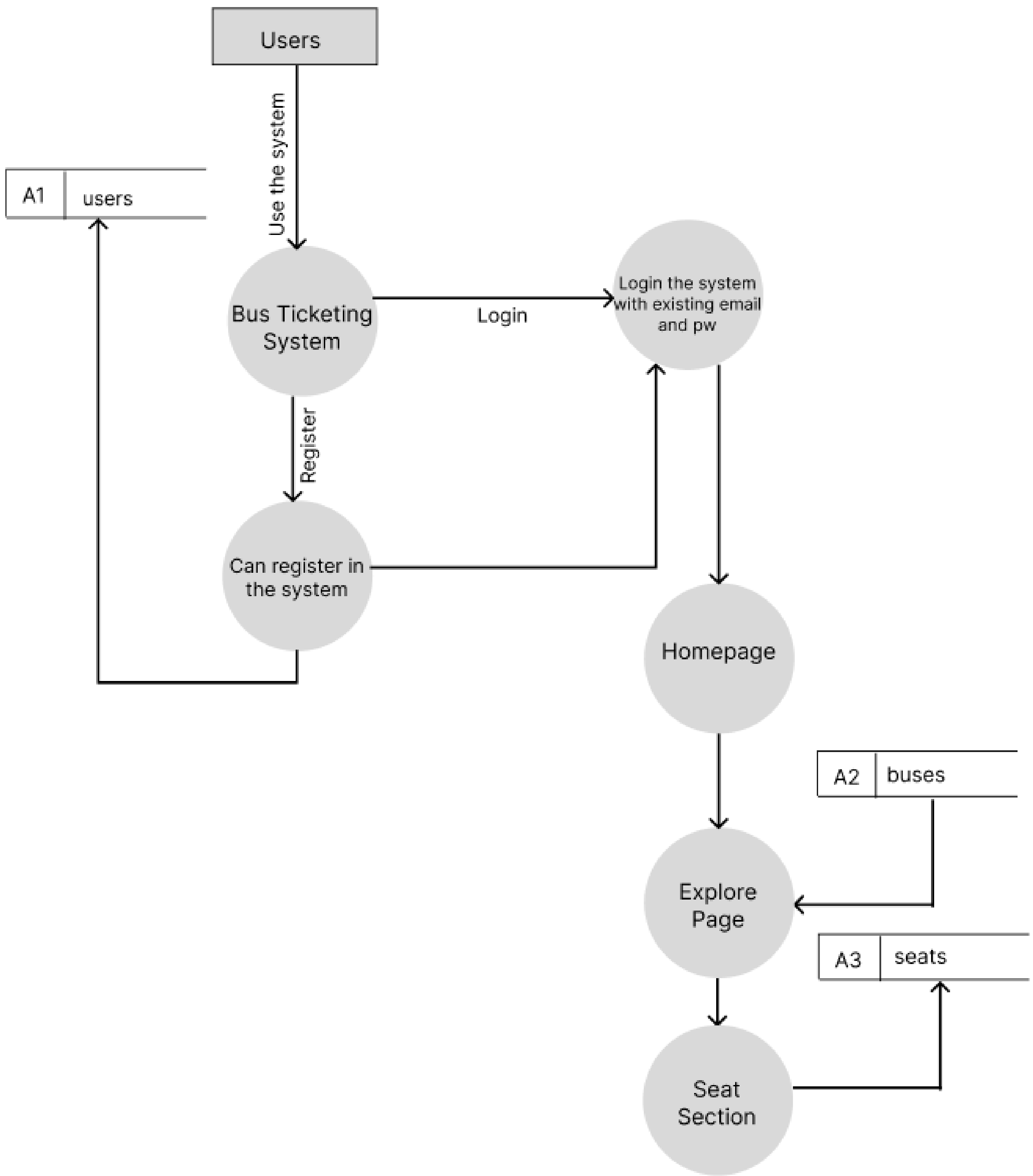

Figure 3. 6 Physical DFD of Bus Ticketing System

The above diagram depicts physical DFD of Bus Ticketing System. User can login or register to the website. The registered data is stored in users table in database. Similarly, a user can browse home page, explore and book the seats. All the data related to specific section are stored in specific table.

3.2 System Design

3.2.1 Architectural Design

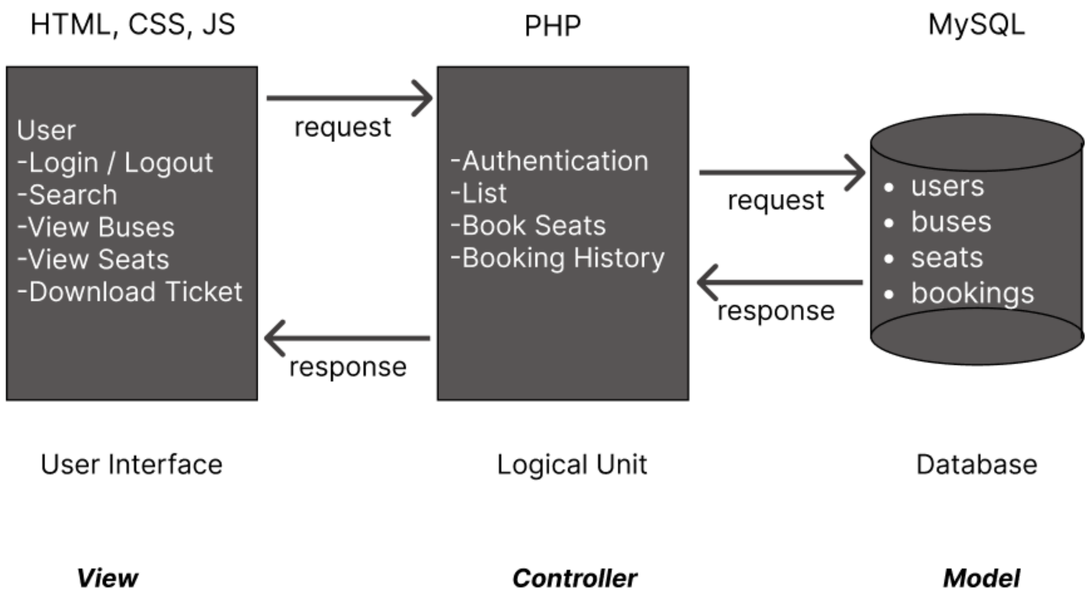

Figure 3. 7 Architectural Design of Bus Ticketing System

In above architecture design, the main element in charge of processing the data gathered from the various input systems is the system server. User can explore the system but for booking user need to register first. After authentication, user can view buses and book the seats. The booked ticket can be also download on the user’s device. Each user's data is stored in a database.

3.2.2 Database Schema Design

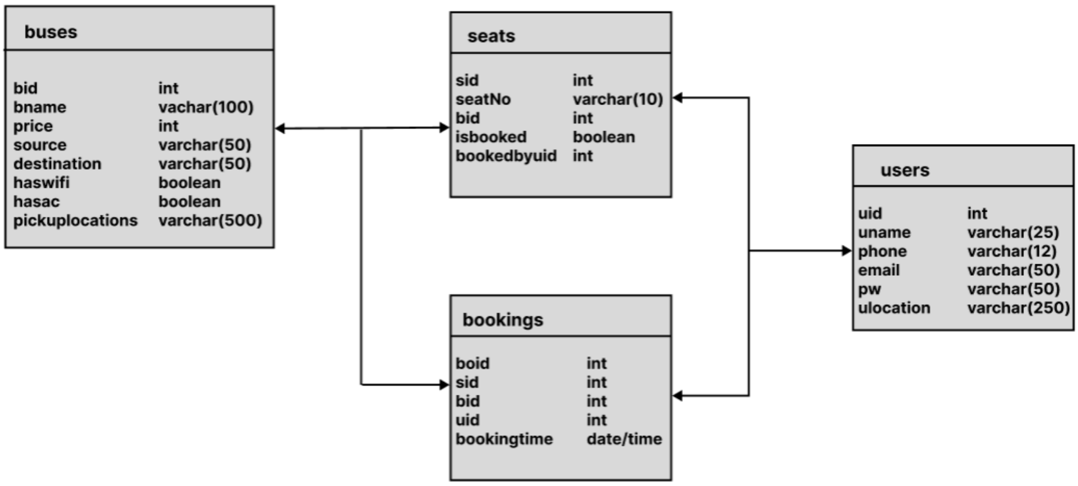

Figure 3. 8 Database Schema Design of Bus Ticketing System

The above diagram is database schema design of Bus Ticketing System. There are four tables: users, buses, seats and bookings. Every table has a unique key. Similarly, the tables also use foreign keys. The database follows normalization principles.

3.2.3 User Interface Design

Figure 3. 9 Login Page of Bus Ticketing System

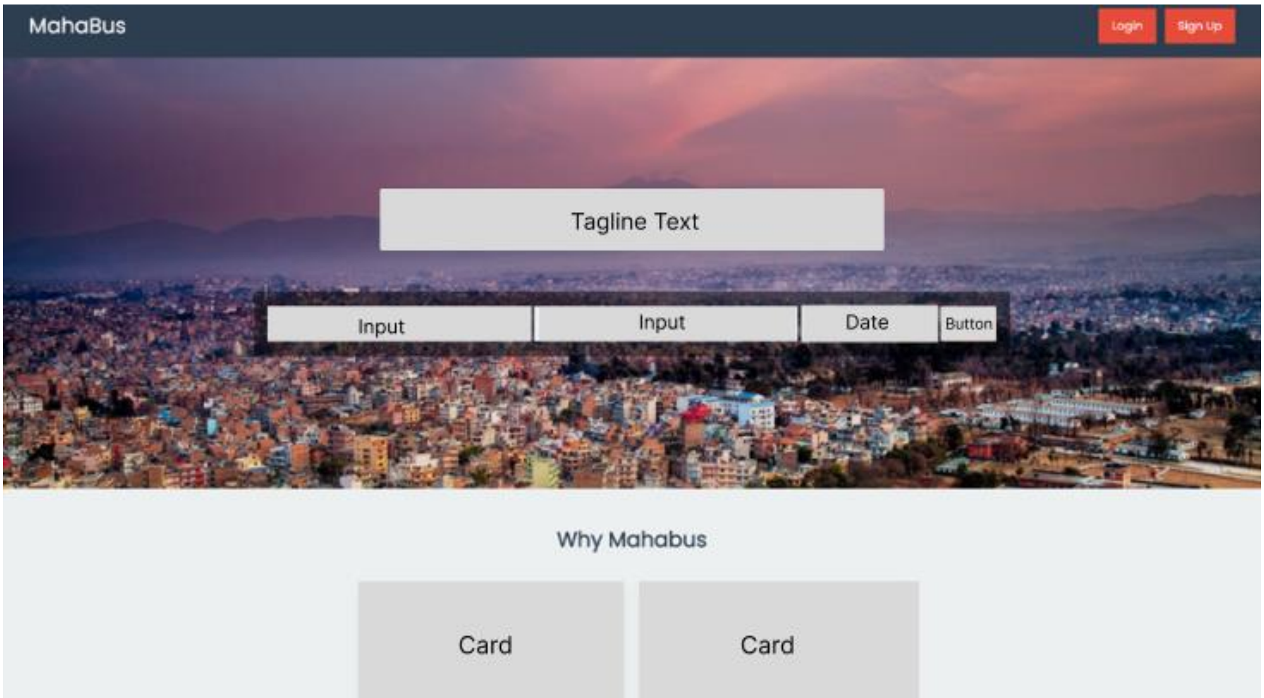

Figure 3. 10 Home Page of Bus Ticketing System

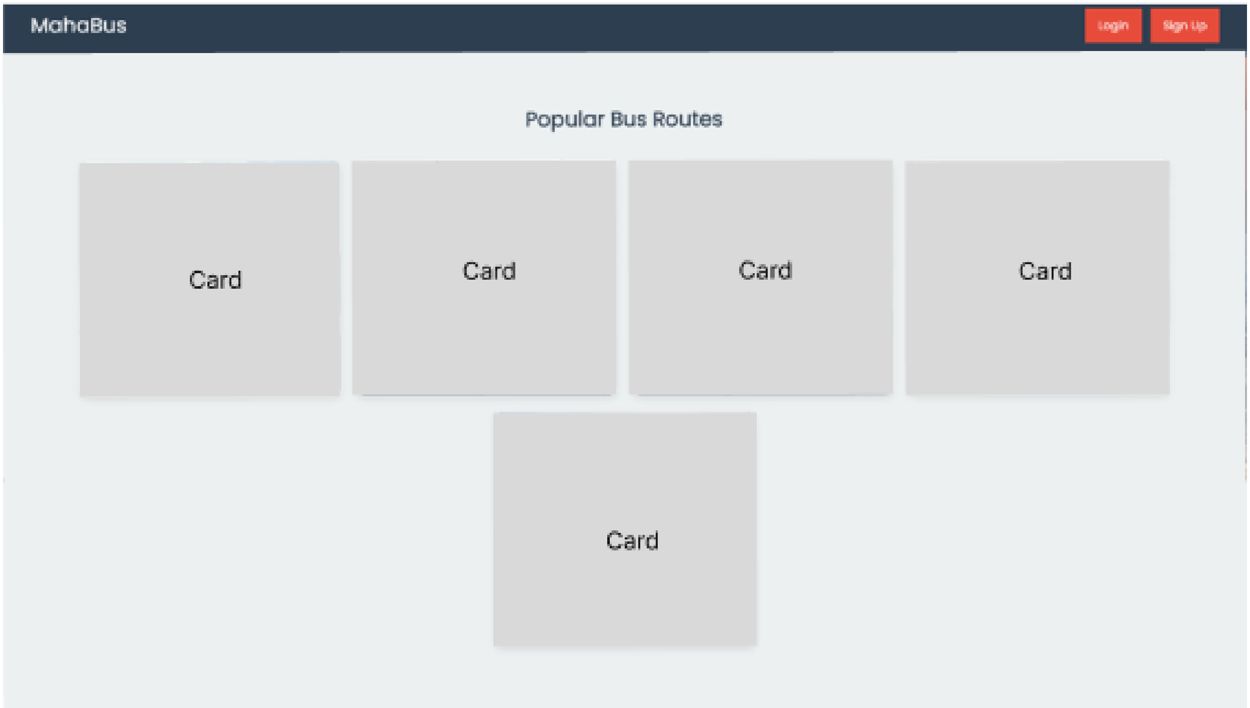

Figure 3. 11 Explore Page of Bus Ticketing System

Figure 3. 12 Seats Detail Page of Bus Ticketing System

CHAPTER 4

IMPLEMENTATION AND TESTING

4.1 Implementation

Data was gathered in the initial phase. Data collection takes longer than the other phases. The entire project's physical design has been translated into functional computer code. The chapter before addressed a variety of techniques and technologies that were used to construct the system.

4.1.1 Tools Used

The tools that are used while designing this web application are:

Frontend: HTML, CSS, JavaScript
Backend: PHP
Database: MySQL
Documentation tools: Microsoft Word, Microsoft Excel, Microsoft PowerPoint

1. **Visual Studio Code(VS Code)**

Visual Studio Code (VS Code) is a popular source code editor that supports HTML and CSS development. VS Code provides syntax highlighting, auto-completion, and code formatting for HTML files. We can create and edit HTML documents using the built-in HTML language. We can write CSS code and apply styles to HTML elements. We can use CSS to define the visual appearance of our web pages, including colors, fonts, layout, and more. VS Code provides debugging capabilities for JavaScript, which is commonly used to enhance interactivity and functionality in web development. [16]

2. **HTML(Hypertext Markup Language) & CSS(Cascading Style Sheet)**

In this project, we use HTML and CSS for web development. These are essential technologies for web development. They work together to create the structure, layout, and visual presentation of web pages. HTML is used to create the basic structure and elements of a web page. It defines the headings, paragraphs, lists, tables, forms, images, and other content on the page. And CSS enables you to build responsive websites that adapt and scale based on the device or screen size. By using CSS media queries, you can apply different styles to different devices, making your site mobile-friendly and ensuring a good user experience across various devices. [17]

3. JavaScript(JS)

In this project, we made the use of JavaScript to add interactivity to webpages and also for some client side form validation. JS is a versatile programming language that is primarily used for adding interactivity and dynamic functionality to websites. By transforming a static web page into an interactive one, JS augmentation enhances the user experience of the website. Just to review, JS gives web pages behavior. [18]

4. PHP

PHP (Hypertext Preprocessor) is used extensively in this project for backend programming or server side scripting. It is also used to integrate and connect SQL queries. It is a server-side scripting language used for web development. It is designed to interact with databases. It enables you to build applications that store and retrieve data dynamically. We can use PHP to retrieve data from databases, insert or update data, and perform queries to fetch specific information. [19]

5. MySQL

In this project, we used MySQL as to store data in the form of various tables such as user table, property table and photos table. MySQL is an open-source relational database management system (RDBMS) that is used to store, manage, and retrieve structured data. It allows you to create and organize structured data in tables, with each table consisting of rows and columns. It enables you to retrieve data from the database using SQL (Structured Query Language) queries. [20]

6. XAMPP Server

We utilized XAMPP Server to host our project locally. It acted as a host for the PHP website as well as MySQL database. XAMPP is a software package that simplifies the setup and deployment of a local web server environment. It provides the necessary tools and services to develop and test websites locally before deploying them to a live server. [21]

7. Microsoft Word

Microsoft Word is a computer program that helps you create and edit documents. It's like a digital notepad where you can type letters, stories, reports, or anything you want to write. In this project, we used it for documentation purposes and report making purpose. You can change the text's appearance, like making it bold or colorful, and add pictures to your documents. It's handy for making things look neat and organized before you print them or share them with others. [22]

8. Microsoft Excel

Microsoft Excel is like a digital spreadsheet. We used excel to record timeline of the project in the form of Gantt chart. It helps you organize and calculate numbers. Imagine a table where you can put numbers and do math with them easily. You can make lists, budgets, or charts to show information in a clear way. It's great for managing finances, keeping track of data, and making things like graphs and tables.

9. Microsoft PowerPoint

Microsoft PowerPoint is like a digital tool for making presentations. It helps you create slides that show pictures, text, and even videos to explain or share something. It's like making a picture story to tell people about a topic. You can make each slide look nice and add transitions to make the presentation interesting. In our project, we used PowerPoint to create presentations.

4.1.2 Implementation details of modules (Description of Procedures/Functions)

a. Registration Module

When creating an account, the form's fields for name, email and password must all be filled out. The user is required to enter all the necessary information when filling out the input box. After filling out registration form when creating an account, user data is kept in the database. The technology allows users to log in once an account has been successfully created. While registering, the user needs to compulsorily provide username, email and password and phone number. After the user is registered, they can further process to log in.

b. Login Module

After successfully creating an account, a user can log in to the system using the login module. It has two fields, including an email and password field. Only user can access when the email address and password match with those in the database. An error message is displayed if the user enters the invalid email address and password.

c. List Buses Module

After logging into the system, the user will be able to access the List Buses module. In this module, the user can search for available buses by entering relevant criteria such as source and destination locations. The system will then display a list of buses that match the entered criteria.

For each listed bus, details such as price and available amenities (e.g., Wi-Fi, air conditioning) will be shown. The user can then select a bus to proceed. All this information is pulled from the database and displayed in real time.

d. View Seats Module

Once a user selects a bus from the List Buses module, they can access the View Seats module. This module provides the user with a visual representation of the seating layout of the selected bus.

Seats that are already booked will be marked as unavailable (often shown in grey), while available seats will be selectable by the user. The seat availability information is retrieved from the database and displayed in real time. If there are no available seats for the selected bus, an error message will inform the user.

e. Book Seat Module

After selecting a seat from the View Seats module, the user is redirected to the Book Seat module. In this module, the selected seat and bus details (e.g., seat number, bus route, departure) are displayed. The user will need to confirm their booking by reviewing the details. The displayed ticket can be downloaded also on user's device.

4.1.3 Algorithms

1. Closest Pickup Location Algorithm

Breakdown of the Algorithm:

Distance Calculation: It calculates the distance between the user's location and a list of coordinates (potential pickup locations) using the Haversine formula.

Reverse Geocoding: Once the closest location is found, it uses a reverse geocoding service to convert the coordinates of that location into a human-readable place name.

Finding the Best Pickup: It identifies the nearest pickup location by comparing distances and returns the corresponding place name.

This algorithm is useful for applications like a bus ticketing system, where you need to suggest the nearest bus stops or pickup points based on the user's location. [23]

```
1. FUNCTION degToRad(deg)
   a. RETURN (deg * π) / 180

2. FUNCTION findDistance(lat1, lon1, lat2, lon2)
   a. SET R = 6371 // Earth's radius in kilometers
   b. SET dLat = degToRad(lat2 - lat1)
   c. SET dLon = degToRad(lon2 - lon1)
   d. SET a = sin(dLat / 2)² + cos(degToRad(lat1)) * cos(degToRad(lat2)) * sin(dLon / 2)²
   e. SET c = 2 * atan2(sqrt(a), sqrt(1 - a))
   f. SET d = R * c // Distance in kilometers
   g. RETURN d

3. FUNCTION getPlaceNameFromCoordinates(latitude, longitude)
   a. SET url = "https://nominatim.openstreetmap.org/reverse?format=json&lat=latitude&lon=longitude&accept-
language=en"
   b. SET options = { "http": { "header": "User-Agent: YourApp/1.0" } }
   c. CREATE context using options
   d. SET response = GET(url, context)
   e. IF response IS NULL THEN
      i. RETURN "Failed to get a response from the API"
   f. SET responseData = parseJSON(response)
   g. IF responseData contains 'display_name' THEN
      i. RETURN responseData['display_name']
   h. ELSE
      i. RETURN "Place not found"

4. FUNCTION bestPickup(ulocation, coordsArray)
   a. CREATE empty list distances
   b. CREATE empty list corrCoords
   c. FOR each coords IN coordsArray DO
      i. SET distance = findDistance(ulocation[0], ulocation[1], coords[0], coords[1])
      ii. ADD distance to distances
      iii. ADD coords to corrCoords
   d. SET minDistance = minimum value in distances
   e. SET index = index of minDistance in distances
   f. RETURN getPlaceNameFromCoordinates(corrCoords[index][0], corrCoords[index][1])
```

Figure 4. 1 Pseudo Code for Closest Pickup Location Algorithm

This algorithm is composed of several functions that calculate distances, convert coordinates, and identify the best pickup point based on a user's location. The degToRad function converts degrees to radians, which is essential for geographic calculations involving the Earth's curvature. The findDistance function implements the Haversine formula to calculate the shortest distance between two points on Earth given their latitude and longitude.

The getPlaceNameFromCoordinates function uses the Nominatim API to fetch the name of a place based on its latitude and longitude. It constructs a request URL with the coordinates, sends the request using an HTTP context with a custom user-agent, and parses the response. If successful, it retrieves the place's display name from the API response; otherwise, it returns an error message or "Place not found."

The bestPickup function finds the closest coordinate to a user's location from an array of coordinates. It calculates the distances of all points using findDistance, identifies the minimum distance, and determines the corresponding point.

Finally, it calls getPlaceNameFromCoordinates to fetch the human-readable name of the closest location, rounding the coordinates for precision. This is useful for optimizing logistics or suggesting nearby locations.

2. User Preference Analysis and Bus Sorting Algorithm Breakdown of the Algorithm:

Breakdown of the Algorithm:

User Preference Analysis: It calculates the average price of bookings and determines whether the user prefers buses with Wi-Fi and air conditioning based on their booking history.

Bus Sorting by Preferences: It sorts a list of buses based on the user's preferences, prioritizing those that are closer to the average price, followed by availability of Wi-Fi and AC, and finally by bus name. [24]

```
FUNCTION analyzeUserPreferences(bookings)
   SET totalPrice = 0
   SET count = count(bookings)
   SET wifiCount = 0
   SET acCount = 0

   FOR each booking IN bookings DO
      totalPrice += booking['price']
      IF booking['haswifi'] THEN
         wifiCount++
      IF booking['hasac'] THEN
         acCount++

   SET averagePrice = count > 0 ? totalPrice / count : 0
   SET wifiPreference = wifiCount > count / 2
   SET acPreference = acCount > count / 2

   RETURN {
      'averagePrice': averagePrice,
      'wifiPreference': wifiPreference,
      'acPreference': acPreference
   }

FUNCTION sortBusesByUserPreferences(buses, preferences)
   usort(buses, FUNCTION(a, b) USE (preferences)
      SET priceDiffA = abs(preferences['averagePrice'] - a['price'])
      SET priceDiffB = abs(preferences['averagePrice'] - b['price'])

      IF priceDiffA != priceDiffB THEN
         RETURN priceDiffA <=> priceDiffB
      IF a['haswifi'] != b['haswifi'] THEN
         RETURN (a['haswifi'] === true) ? -1 : 1
      IF a['hasac'] != b['hasac'] THEN
         RETURN (a['hasac'] === true) ? -1 : 1

      RETURN strcmp(a['bname'], b['bname'])
   )

   RETURN buses
```

Figure 4. 2 Pseudo Code for Bus Sorting Algorithm

The analyzeUserPreferences function processes a user's past bookings to extract insights. It calculates the average price of booked buses, as well as the user's preference for Wi-Fi and air conditioning (AC) based on the count of buses with these features. Wi-Fi or AC is marked as preferred if more than half of the bookings include it. This information provides a baseline for understanding the user's needs.

The sortBusesByUserPreferences function organizes buses according to the user's preferences. It prioritizes buses with ticket prices closer to the user's average spending. If two buses have the same price difference, Wi-Fi and AC preferences are considered, giving priority to buses with these amenities. If all else is equal, the buses are sorted alphabetically by name, ensuring a consistent ranking order.

The contentBased function combines the above functionalities to recommend buses tailored to the user's preferences. It first analyzes past bookings to understand user behavior, then sorts the available buses using the derived preferences. The result is a list of buses ranked in order of suitability, helping users find options that best match their past choices and needs.

4.2 Testing

The different training and testing datasets are used to test the system. The purpose of this test is to determine whether or not the system is delivering the correct summaries. Our system is continually tested since it is in the development phase. Following are the tests that were run:

Table 4. 1 Test Cases for Unit Testing

Test case ID	Test Scenario	Test Steps	Test Data	Expected Result	Pass/ Fail
T1	User Register new account	Email: shyam@gmail.com Password: shyam	Redirect to success page	As expected	Pass
T2	User enters wrong email and password	Email: shyamm@gmail.com Password: shyaam	Redirect to error page	As expected	Pass
T3	User enters right email and password	Email: shyam@gmail.com Password: shyam	Redirect to home page	As expected	Pass

Table 4. 2 Search Buses

Test case ID	Test Scenario	Test Steps	Test Data	Expected Result	Pass/ Fail
T1	Search buses with pickup and destination	Pickup Destination: Kathmandu To Destination: Pokhara	List the available bus data	As expected	Pass
T2	Search buses with wrong pickup and destination	Pickup Destination: Kathmandu To Destination: Delhi	No tickets found	As expected	Pass

Table 4. 3 Book a Seat

Test case ID	Test Scenario	Test Steps	Test Data	Expected Result	Pass/ Fail
T1	Book seat/s while logged in	Login and book the seats	The seat/s gets booked and updated to the database	As expected	Pass
T2	Book seat/s while logged out	Book the seat/s while logged out	The book button is not displayed to the user	As expected	Pass

Test Cases for System Testing

In the following table, different test cases have been tested in following:

Table 4. 4 Test Plan for System Testing

S.N	Test Plan
1	To check if the user registration module works properly.

2	To check if the user login module works properly.
3	To check if the admin upload module works properly.
4	To check if the search module works properly.

Table 4. 5 Test Case 1 for Registration Module

Test no.	1
Test case	To check if user is created or not
Expected result	Users should be created.
Actual result	User was created.
Conclusion	Expected and actual results matched.

Table 4. 6 Test Case 2 for Login Module

Test no.	2
Test case	To check if the login module works.
Expected result	Users should be redirected to the Dashboard.
Actual result	User redirected to the Dashboard.
Conclusion	Expected and actual results matched.

Table 4. 7 Test Case 3 for Admin Upload Module

Test no.	3
Test case	To check if upload module works
Expected result	Uploaded data should be added.

Actual result	Uploaded data added.
Conclusion	Expected and actual results matched.

Table 4. 8 Test Case 4 for Search Module

Test no.	4
Test case	To check if the search module works.
Expected result	Buses should be listed according to the search address.
Actual result	Buses listed according to search address.
Conclusion	Expected and actual results matched.

Test Cases for Algorithm

In the following table, different test cases have been tested in following:

Table 4. 9 Test Case for Closest Pickup Algorithm

Test no.	1
Test case	To check if closest pickup location is suggested.
Expected result	Closest pickup location should be suggested.
Actual result	Closest pickup location is suggested.
Conclusion	Expected and actual results matched.

Table 4. 10 Test Case for Content Base Algorithm

Test no.	1
Test case	To check if buses are suggested according to the user preferences.
Expected result	Buses should be suggested according to the user preferences.
Actual result	Buses suggested according to the user preferences.
Conclusion	Expected and actual results matched.

CHAPTER 5

CONCLUSION AND FUTURE RECOMMENDATION

5.1 Lesson Learnt/Outcome

5.1.1 Lesson Learnt

This project's completion made it feasible to accomplish the project's objective. The user can view & search the seats through a web browser after completing the registration form. Every project forces us to grow and learn in a variety of ways. With this project, we have mastered the problem solving skills independently, writing skills, making correct use of instructions, communicating effectively, time management, and team leadership. Most importantly, we have learnt version control tools and PHP programming.

5.1.1.1 Soft Skills

The soft skills we acquired include carrying out the mission, setting goals, displaying self-assurance through effective communication, having a positive outlook, participating in teamwork, problem-solving abilities, writing abilities, correctly using instructions, time management, and team leadership.

5.1.1.2 Hard Skills

The process of working on this project often involved the development of various talents. First and foremost, knowledge of web development languages like HTML, CSS, and JavaScript was necessary. These coding languages made it possible to create interactive and aesthetically pleasing web pages. Additionally, for the implementation of dynamic functionality and database management, proficiency in the backend programming language PHP was essential. To effectively address technical challenges, additional crucial hard skills include proficiency in debugging and troubleshooting. The ability to master these hard skills gives people the resources they need to successfully create, manage, and optimize websites.

5.1.2 Outcomes

In the project that follows, we have mastered the problem solving skills independently, writing skills, making correct use of instructions, communicating effectively, time management, and team leadership

5.2 Conclusion

After the successful completion of the project, the user will be able to view & search from anywhere with a simple login in our website. This project is based in UI/UX Design so the users can easily use it. This project has been developed in PHP and database keeping in mind the specifications of the system has been created in My SQL server. By integrating real-time seat availability, multiple payment methods, and a dynamic search system, the platform will meet the needs of both passengers and bus operators, enhancing overall efficiency and customer satisfaction. The authors have achieved the objectives of this project work.

5.3 Future Recommendation

Future additions to this website can enhance several aspects, including portability and user experience. There is still work to be done, so this application might be thought of as the beginning of something much bigger. While each of these will need more time and resources to fulfill, they are all still very attainable and realistic objectives.

- Greater User Experience
- Add forget password options
- Add online payment system
- OTP (One Time Password) feature can be added.

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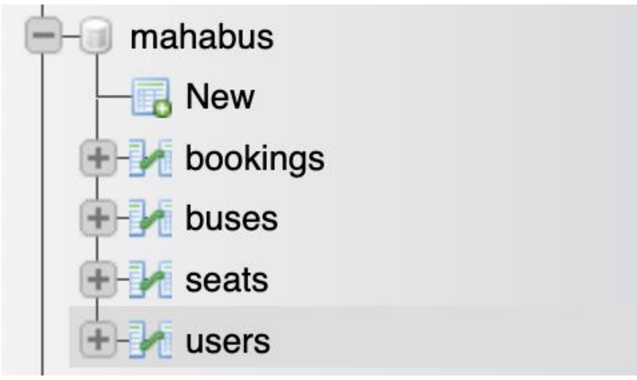
APPENDIX

System Screenshots:

Database Overviews:



1.Users Table

←T→	id	uname	email	phone	pw	ulocation
Edit Copy Delete	6	Rubek Maharjan	rubek@gmail.com	9745706122	fa8f9bd434e4fef76b505f4a90d8e026	27.67246339187131zzz85.33831797516292
Edit Copy Delete	7	admin	admin@mahabus.com	9745796123	21232f297a57a5a743894a0e4a801fc3	NULL
Edit Copy Delete	8	Stephen Shrestha	stephen@gmail.com	9813158331	fe5e5085144feef43835113f45d726de	NULL

2. Seats Table

←T→	sid	bid	seatno	isbooked	bookedbyuid
Edit Copy Delete	211	13	A1	0	NULL
Edit Copy Delete	212	13	A2	0	NULL
Edit Copy Delete	213	13	A3	0	NULL
Edit Copy Delete	214	13	A4	0	NULL
Edit Copy Delete	215	13	A5	0	NULL
Edit Copy Delete	216	13	A6	0	NULL
Edit Copy Delete	217	13	A7	0	NULL
Edit Copy Delete	218	13	A8	0	NULL
Edit Copy Delete	219	13	A9	0	NULL
Edit Copy Delete	220	13	A10	0	NULL
Edit Copy Delete	221	13	A11	0	NULL
Edit Copy Delete	222	13	B1	0	NULL
Edit Copy Delete	223	13	B2	0	NULL
Edit Copy Delete	224	13	B3	0	NULL
Edit Copy Delete	225	13	B4	0	NULL
Edit Copy Delete	226	13	B5	0	NULL

3. Buses Table

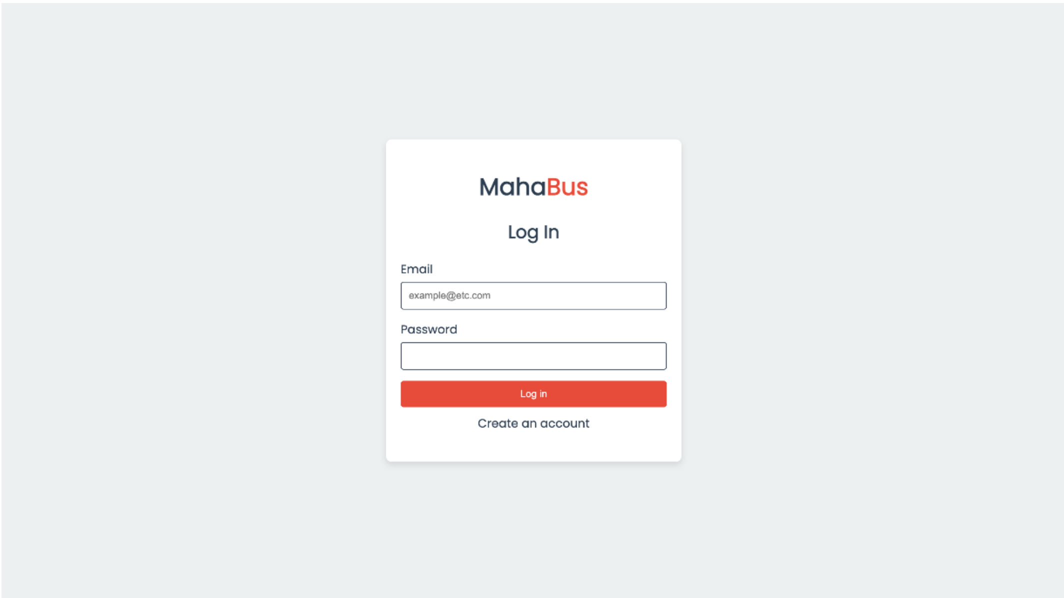
←T→			bid	bname	price	haswifi	hasac	source	destination	pickuplocations
Edit	Copy	Delete	13	Mahalaxmi Yatayat	1350	0	1	Kathmandu	Janakpur	27.674974764923384zzz85.29261320829393xxx27.660379...
Edit	Copy	Delete	14	Helux Yatayat	1700	1	1	Kathmandu	Surkhet	27.70602336785347zzz85.28137207031251xxx27.6695421...
Edit	Copy	Delete	15	Night Yatayat	1200	1	1	Kathmandu	Pokhara	27.69280035030002zzz85.27999877929689xxx27.7250191...
Edit	Copy	Delete	16	Lalima Yatayat	1500	1	1	Kathmandu	Jhapa	27.665187863577245zzz85.42429357767107xxx27.691942...
Edit	Copy	Delete	17	Bhimsen Yatayat	1500	1	1	Kathmandu	Dolakha	27.74028639349318zzz85.33810950815679xxx27.7056404...

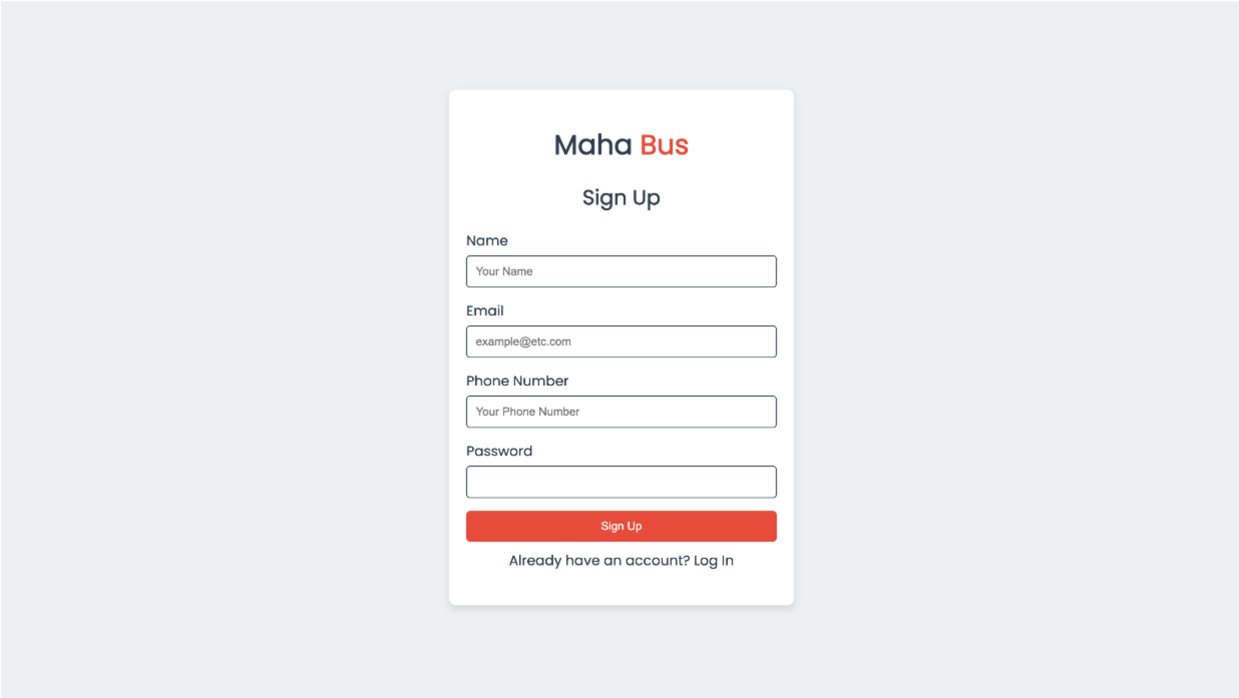
4. Bookings Table

←T→			boid	bid	sid	uid	bookingtime
Edit	Copy	Delete	29	15	261	6	2024-09-21 23:37:07
Edit	Copy	Delete	30	15	262	6	2024-09-21 23:37:07
Edit	Copy	Delete	31	15	268	6	2024-09-22 08:05:06
Edit	Copy	Delete	32	15	271	6	2024-09-22 08:27:26
Edit	Copy	Delete	33	13	220	6	2024-09-22 09:46:48
Edit	Copy	Delete	34	13	221	6	2024-09-22 09:46:48
Edit	Copy	Delete	35	18	333	6	2024-09-22 09:48:49

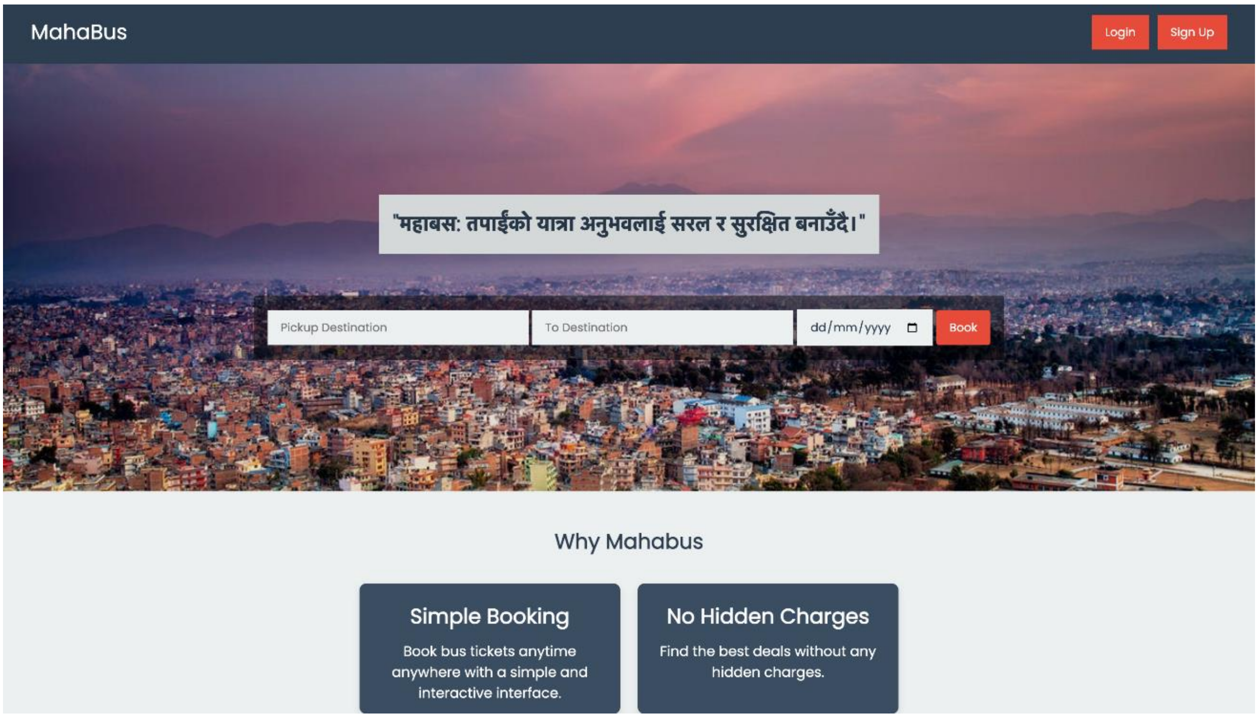
User Interface:

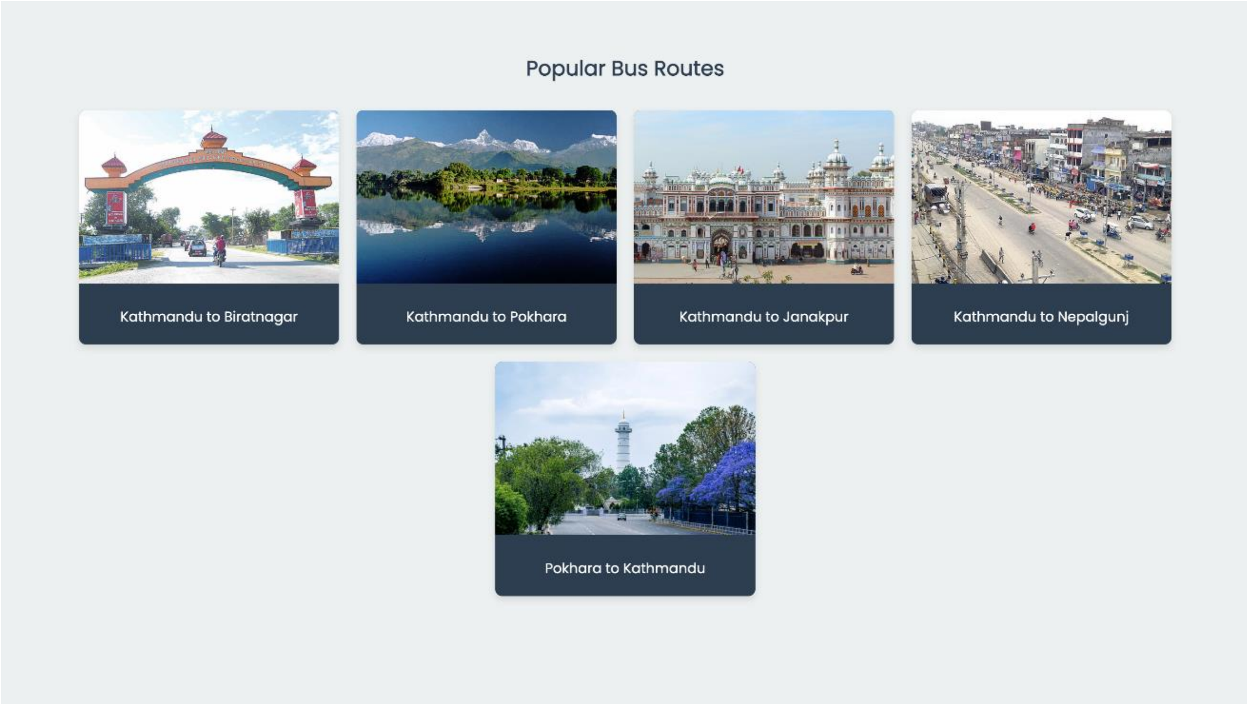
1.Signup/ Signin




Maha Bus
Sign Up
Name
Your Name
Email
example@etc.com
Phone Number
Your Phone Number
Password
Sign Up
Already have an account? Log In

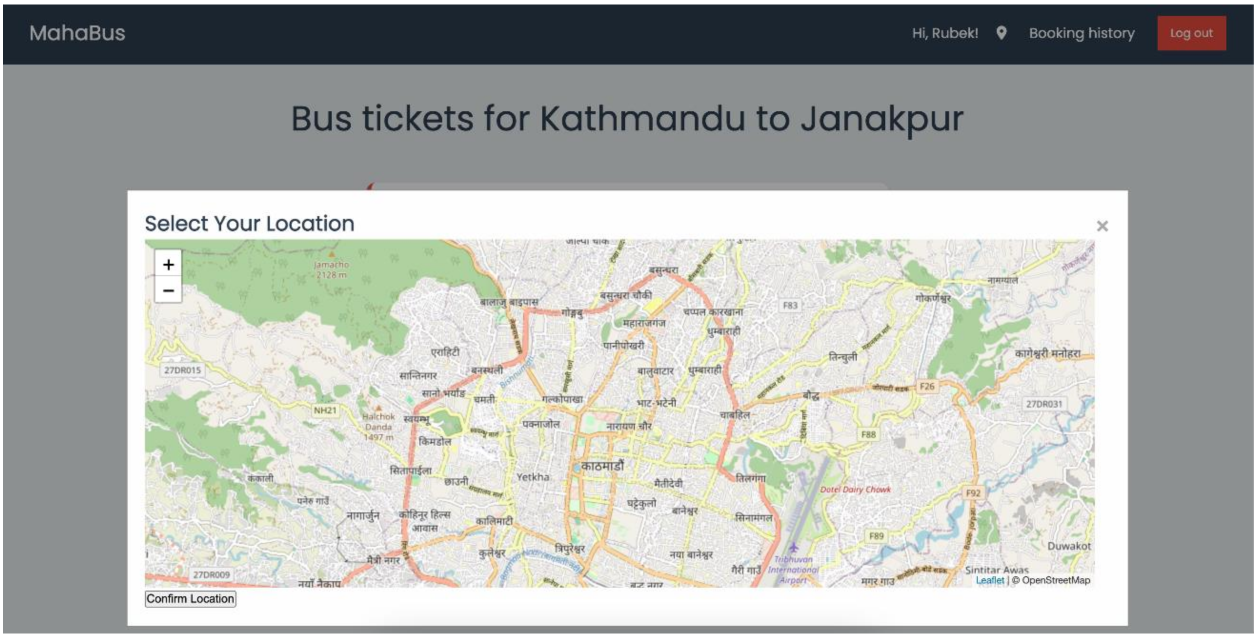
2. Home Page


MahaBus
Login
Sign Up
"महाबस: तपाईंको यात्रा अनुभवलाई सरल र सुरक्षित बनाउँदै।"
Pickup Destination
To Destination
dd/mm/yyyy
Book
Why Mahabus
Simple Booking
Book bus tickets anytime anywhere with a simple and interactive interface.
No Hidden Charges
Find the best deals without any hidden charges.

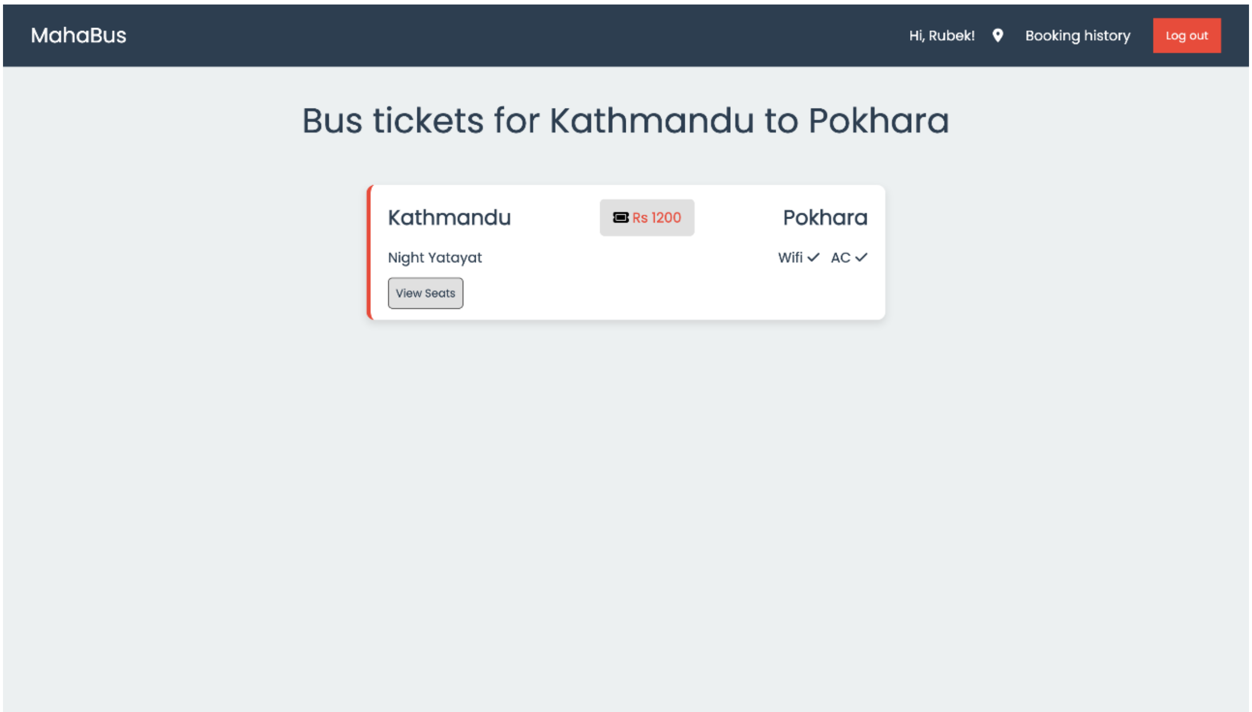


3. Book Seats

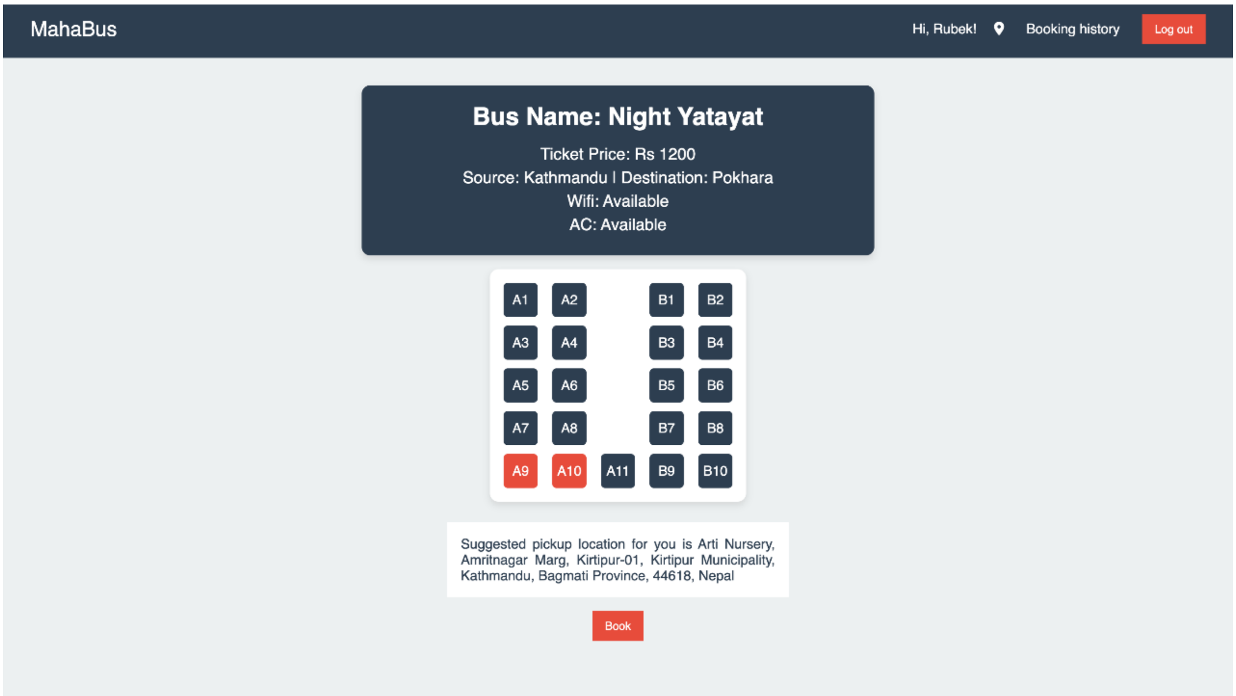
a. Get User Location



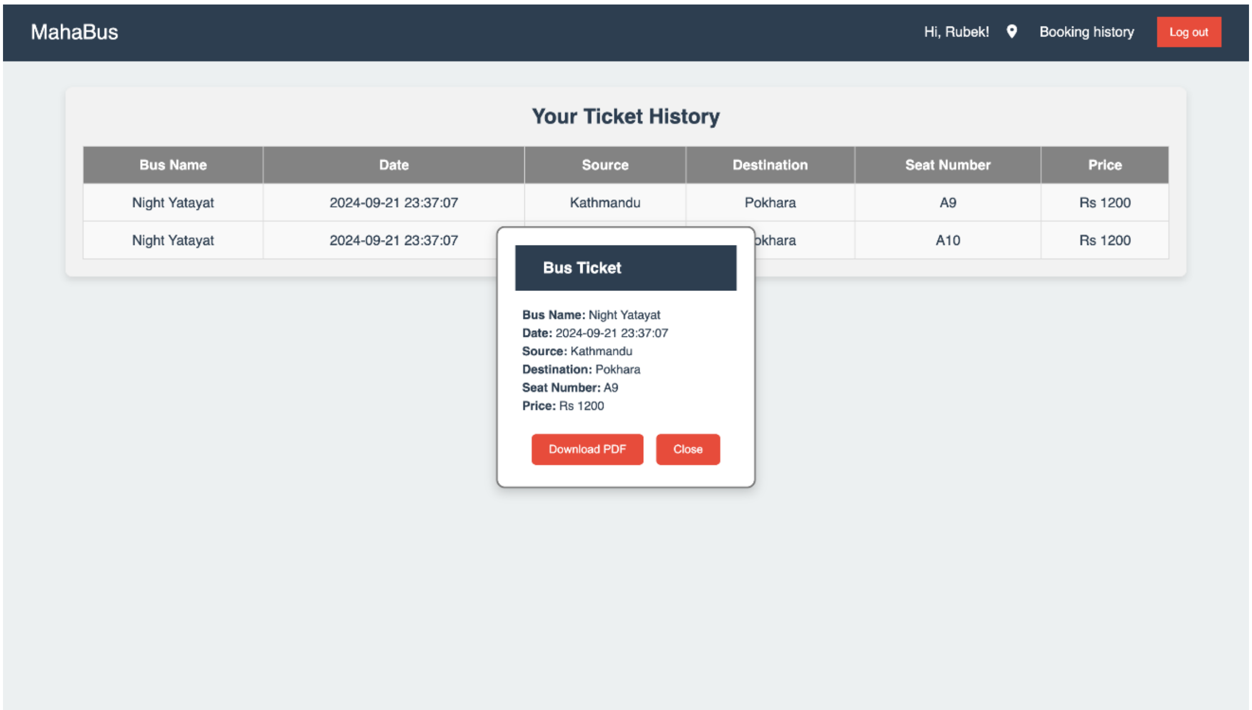
b.Choose a Bus



c. Book Seats



d. View Ticket



4. Admin Panel